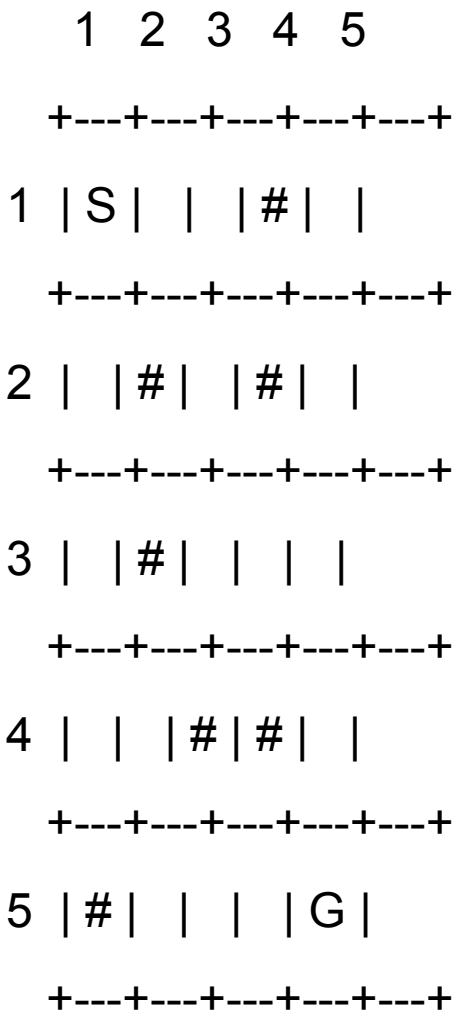
2. 5×5 Grid



= Obstacle

Manhattan Distance

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

Best-First Search always expands the node having the **lowest heuristic value h(n)**.

Step-by-step Node Expansion

Step	Expanded Node	h(n)	Priority Queue
1	S(1,1)	8	(2,1): 7
2	(2,1)	7	(3,1): 6
3	(3,1)	6	(4,1): 5
4	(4,1)	5	(4,2): 4
5	(4,2)	4	(5,2): 3
6	(5,2)	3	(5,3): 2
7	(5,3)	2	(5,4): 1
8	(5,4)	1	(5,5): 0
9	G(5,5)	0	Empty

Priority Queue Updates

Initial:

[S(1,1)]

After expanding S:

[(2,1)]

After expanding (2,1):

[(3,1)]

After expanding (3,1):

[(4,1)]

After expanding (4,1):

[(4,2)]

After expanding (4,2):

[(5,2)]

After expanding (5,2):

[(5,3)]

After expanding (5,3):

[(5,4)]

After expanding (5,4):

[G]

Final Path

S

↓

(2,1)

↓

(3,1)

↓

(4,1) → (4,2)

↓

(5,2) → (5,3) → (5,4) → G

Therefore,

$$S \rightarrow (1) \rightarrow (1) \rightarrow (1) \rightarrow (2) \rightarrow (2) \rightarrow (3) \rightarrow (4) \rightarrow G$$

Number of moves = **8**

Cost of each move = **1**

$$Total\ Cost = 8$$

Answer: The robot reaches the goal with an **8-move path** and the total path cost is **8**.

7. **7. Ambulance Routing During Floods – A* Search**

Given

Road **B–D is closed**, so it **cannot be used**.

A* uses:

$$f(n) = g(n) + h(n)$$

where:

- **g(n)** = actual cost from S to node

- **h(n)** = heuristic estimate to G
- **f(n)** = total estimated cost

---

i. A* Search with B–D Closed

Step 1: Start at S

For S:

$$g(S) = 0,\ h(S) = 6,\ f(S) = 0 + 6 = 6$$

Open list:

Node	g	h	f
S	0	6	6

Closed list: Empty

Expand **S**.

Neighbors:

- A: $g = 3$
- B: $g = 2$

Step 2: Expand S

Node	g	h	f
A	3	6	9
B	2	4	6

Open: A(3,6,9), B(2,4,6)
Closed: S

Choose **B** because it has the lowest f = 6.

Step 3: Expand B

B → C is available.

B → D is **closed**, so it is ignored.

For C:

$$g(C) = g(B) + 1 = 2 + 1 = 3 f(C) = 3 + 4 = 7$$

Open:

Node	g	h	f
C	3	4	7
A	3	6	9

Closed: S, B

Choose **C**.

Step 4: Expand C

C has two available routes:

- C → D = 2
- C → E = 3

For D:

$$g(D) = 3 + 2 = 5 f(D) = 5 + 2 = 7$$

For E:

$$g(E) = 3 + 3 = 6 f(E) = 6 + 3 = 9$$

Open:

Node	g	h	f
D	5	2	7
A	3	6	9
E	6	3	9

Closed: S, B, C

Choose **D**.

Step 5: Expand D

D → G has cost 2.

$$g(G) = 5 + 2 = 7 f(G) = 7 + 0 = 7$$

Open:

Node	g	h	f
G	7	0	7
A	3	6	9
E	6	3	9

Closed: S, B, C, D

Choose **G**. Goal reached.

---

ii. Route and Total Cost

The route is:

$$S \rightarrow B \rightarrow C \rightarrow D \rightarrow G$$

Cost:

$$S - B = 2B - C = 1C - D = 2D - G = 2$$

Therefore,

$$2 + 1 + 2 + 2 = 7\,minutes$$

Answer

Route: S → B → C → D → G
Total cost: 7 minutes

---

iii. If B–D Were Open

Now B → D with cost **2** is allowed.

After expanding B:

For D:

$$g(D) = g(B) + 2 = 2 + 2 = 4 f(D) = 4 + 2 = 6$$

So the Open list becomes:

Node	g	h	f
D	4	2	6
C	3	4	7
A	3	6	9

A* selects **D** because f = 6.

From D:

$$g(G) = 4 + 2 = 6 f(G) = 6 + 0 = 6$$

Therefore the route is:

$$S \rightarrow B \rightarrow D \rightarrow G$$

Cost:

$$2 + 2 + 2 = 6\, minutes$$

Answer

If B–D is open:

$$S \rightarrow B \rightarrow D \rightarrow G$$

Total cost = 6 minutes

---

iv. Admissibility and Consistency

Admissible Heuristic

A heuristic is **admissible** if:

$$h(n) \leq \ actual\ shortest\ cost\ from\ n\ to\ G$$

For the **flooded network**:

Node	h(n)	Actual shortest cost to G	Admissible?
S	6	7	Yes
A	6	8	Yes
B	4	5	Yes

Node	h(n)	Actual shortest cost to G	Admissible?
C	4	4	Yes
D	2	2	Yes
E	3	4	Yes
G	0	0	Yes

Therefore:

$$Heuristic\ is\ admissible$$

Consistency

A heuristic is consistent if:

$$h(n) \leq c\left(n^{'}\right) + h\left(n^{'}\right)$$

Check the important edges:

- S → A: $6 \leq 3 + 6 = 9$✓
- S → B: $6 \leq 2 + 4 = 6$✓
- A → C: $6 \leq 4 + 4 = 8$✓
- B → C: $4 \leq 1 + 4 = 5$✓
- C → D: $4 \leq 2 + 2 = 4$✓
- C → E: $4 \leq 3 + 3 = 6$✓
- D → G: $2 \leq 2 + 0 = 2$✓
- E → G: $3 \leq 4 + 0 = 4$✓

Hence:

$$Heuristic\ is\ consistent$$

Final Answers

i. A* with B–D closed:

$$S \rightarrow B \rightarrow C \rightarrow D \rightarrow G$$

ii. Total cost:

$$7\ minutes$$

iii. If B–D is open:

$$S \rightarrow B \rightarrow D \rightarrow G$$

Cost:

$$6\ minutes$$

iv.

$$Heuristic\ is\ both\ admissible\ and\ consistent.$$